MIL-S-12514C
9 January 1974
SUPERSEDING
MIL-S-12514B
27 Sept 1965

MILITARY SPECIFICATION

STARTERS, MOTOR; ACROSS-THE-LINE AND REDUCED VOLTAGE TYPES, A-C INDUCTION, 3/4 TO 100 HP, MANUAL AND ELECTRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers general-purpose starters for alternating current induction motors with ratings from 3/4 to 100 hp, inclusive.

1.2 Classification. The starters shall be of the following types, classes, and styles, as specified (see 6.2):

Type I - Across-the-line

Class 1 - Manually operated
Class 2 - Magnetically operated

Style A - Low-voltage release
Style B - Low-voltage protection

Type II - Reduced voltage

Class 1 - Manually operated
Class 2 - Magnetically operated

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 6110

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SPECIFICATIONS

Federal

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Military

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for the Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-W-5086	- Wire, Electric, Hookup and Interconnecting Polyvinyl Chloride-Insulated, Copper or Copper Alloy Conductor.
MIL-E-17555	- Electronic and Electrical Equipment. Accessories, and Repair Parts; Packaging and Packing of.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-810	- Environmental Test Methods.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for propocal shall apply.

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS - Industrial Controls and Systems.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017).

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110).

AIR POLLUTION CONTROL DISTRICT

Rule - 66

(Application for copies should be addressed to the Air Pollution Control District, County of Los Angeles, CA 90001).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

3. REQUIREMENTS

3.1 *Description.* The starters shall be externally operable, nonreversing, air-breaking, enclosed units equipped with low voltage and overload protection. Starters shall conform to NEMA ICS and as specified herein. Starters shall be of the following sizes: 00, 0, 1, 2, 3, 4, and 5, as specified (see 6.2).

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3.2 First article (preproduction model). The supplier shall furnish one or more complete starters as specified (see 6.2) for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce starters that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and unless otherwise specified herein, all examination and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Temperature.

3.4.1 Operation. When tested as specified in 4.5.2.1, starters shall operate as specified herein in any ambient air temperature from plus 125° F. to minus 25° F. and at relative humidities as specified herein.

3.4.2 Storage. The starters shall not be damaged by storage in any ambient air temperature from plus 160° F. to minus 50° F.

3.5 Voltage. When equipped with all accessories and in adjustment, the starters shall operate from 85 to 110 percent of rated line voltage at rated frequency.

3.6 Ratings. The starters shall have ratings as specified in NEMA ICS, and as specified herein.

3.7 Overload protection. Overload protection shall be provided with each starter by a manual-reset, overload relay which shall provide inverse, time limit protection as permitted by NFPA No. 70 and NEMA ICS. Protectors shall be mounted so as to be removable without any parts other than the cover being removed. The normal operation of the overload relay shall be to open all ungrounded line conductors. Starters for all 3-phase motors shall provide over-current sensing elements for all ungrounded line conductors. A list of interchangeable parts for the overload protector shall be provided, and ratings of each part shall be included.

3.8 Contacts. Contact arcing shall be kept to a minimum.

3.9 Wiring. Wiring shall conform to MIL-W-5086, classification as applicable. A wiring diagram shall be provided with each starter.

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3.10 Enclosures. The starter enclosure shall conform to one of the types listed in NEMA ICS, 1-110, as specified (see 6.2).

3.11 Interchangeability. All parts, components, and attachments shall be interchangeable with starters of the same types and classes produced by the same supplier.

3.12 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.12.1, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.12.1 Composition. Composition I shall be used unless composition II must be used in order to comply with local air pollution regulations in the application of the varnish. When composition II is used, the supplier shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 66 - Air Pollution Control District, County of Los Angeles, CA.

3.13 Altitute. Starters shall be capable of operating at 6000 feet or less in accordance with NEMA ICS and as specified herein.

3.14 Dielectric voltage. The dielectric voltage shall be in accordance with NEMA ICS, 1-109.05.

3.15 Identification marking. The starters shall be identified in accordance with MIL-STD-130. The marking shall be applied to the starters on plates conforming to MIL-P-514, type I, style 3, composition C, of grade A, class 1 material.

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3.16 Treatment and painting. The portions of the starters normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.17 Type I. Across-the-line starters shall connect the motor directly across the line.

3.17.1 Class 1. Manually operated starters shall operate by means of a lever.

3.17.2 Class 2. Magnetically operated starters shall operate by means of 2-wire or 3-wire, start-stop, push buttons. The starters shall be reset by an externally operable push button located in the enclosure.

3.17.2.1 Style A. The push button station shall be a maintained contact switch for low-voltage release.

3.17.2.2 Style B. The push button station shall be a momentary contact switch for low-voltage protection.

3.18 Type II. Reduced-voltage starters shall connect the motor to the line through reduced-voltage taps.

3.18.1 Class 1. Manually operated starters shall be operated by a handle. The words "OFF", "START", and "RUN" shall be permanently marked in appropriate position near the handle. A starting sequence shall be provided which will prevent moving the handle to the "RUN" position prior to being in the "START" position.

3.18.2 Class 2. Magnetically operated starters shall operate by means of 2-wire or 3-wire, momentary-contact, push buttons. The starting sequence shall be governed by an accelerating relay having a manually, controllable variable time limit. The time-limit range shall permit acceleration of all motors with which the starter may be used.

3.19 Workmanship. All parts, components, and assemblies of the starters including castings, forgings, molded parts, stamping, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, scale, flux, and other harmful extraneous material. All edges shall be rounded or beveled.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction starter(s) shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection of all preproduction starters.

4.3.2 Tests. The preproduction starter(s) shall be tested as specified in 4.5.2.1 through 4.5.2.6. Failure of any test shall be cause for rejection of all preproduction starters.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level II.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects listed in 4.5.1. AQL shall be 2.5 percent defective.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.2.1 and 4.5.2.2.2. AQL shall be 2.5 percent defective.

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4.5 Inspection procedure.

4.5.1 Examination. Starters shall be examined as specified herein for the following defects:

101. Components missing or not as specified.
102. Wiring diagram not provided.
103. Parts, components, or attachments not interchangeable.
104. Identification marking not as specified.
105. Treatment and painting not as specified.
106. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Temperature.

4.5.2.1.1 Operation.

4.5.2.1.1.1 Hot-dry. Expose the starter to an ambient air temperature of not less than 125° F. at a relative humidity of not greater than 10 percent for not less than 8 consecutive hours. After this period of time, use the starter to start a matched motor (hp, voltage, frequency and phases), with rated voltage at rated frequency. After the motor has accelerated to its rated speed, load the motor to its rated load. The starter shall remain energized and the motor shall run with its rated load for not less than 4 hours. Temperature of coils, windings, overload protectors in the starter shall be measured at the time the motor is started and at 15 minute intervals until the temperature stabilizes in accordance with NEMA ICS, 1-109.04.8. Any malfunction, inability of the starter to energize the motor or overheating above the specified final temperature (ambient temperature plus temperature rise) for the type of insulation used shall constitute failure of this test.

4.5.2.1.1.2 Hot-humid. Expose the starter to an ambient temperature of not less than 95° F. with a relative humidity of not less than 85 percent for not less than 8 consecutive hours. After this period of time use the starter to start a matched motor (hp, voltage, frequency and phases), with rated voltage at rated frequency. After the motor has accelerated to its rated speed, load the motor with its rated load. The starter shall remain energized and the motor shall run with its rated load for not less than 4 hours. Temperature of coils, windings, overload protectors in the starter shall be measured at the time the motor is started and at 15 minute intervals until the temperature stabilizes in accordance with NEMA ICS 1-109.04.8. Any malfunction, inability of the starter to energize the motor, or overheating above the specified final temperature (ambient temperature plus temperature rise) for the type of insulation used shall constitute failure of this test.

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4.5.2.1.1.3 Low temperature. Expose the starter to an ambient air temperature not greater than minus 25° F. with a relative humidity tending towards saturation for not less than 8 consecutive hours. After this period of time, use the starter to start a matched motor (hp, voltage, frequency and phases), with rated voltage at rated frequency. After the motor has accelerated to its rated speed, load the motor with its rated load. The starter shall remain energized and the motor shall run for not less than 4 hours causing a temperature rise. Any malfunction, inability of the starter to energize the motor, or cracking of materials shall constitute failure of this test.

4.5.2.1.2 Storage temperature.

4.5.2.1.2.1 Hot temperature. Expose the starter to an ambient air temperature of not less than 160° F. for not less than 3 consecutive days. At the end of this period, the starter shall be allowed to reach room temperature. When the starter is within 5° F. of room temperature, use it to start a matched motor (hp, voltage, frequency and phases), with rated voltage at rated frequency. Any malfunction or inability of the starter to energize the motor shall constitute failure of this test.

4.5.2.1.2.2 Cold temperature. Store the starter in an ambient air temperature of not greater than minus 50° F. for not less than 3 consecutive days. At the end of this period the starter shall be allowed to reach room temperature. When the starter is within 5° F. of room temperature, use it to start a matched motor (hp, voltage, frequency and phases), with rated voltage at rated frequency. Any malfunction, inability of the starter to energize the motor, or cracking of materials shall constitute failure of this test.

4.5.2.2 Voltage.

4.5.2.2.1 85 percent voltage. Connect the starter to a supply voltage of not greater than 85 percent of rated voltage at rated frequency. Operate the starter without the motor. Failure of the starter to operate or inability of the contacts to fully close shall constitute failure of this test.

4.5.2.2.2 110 percent voltage. Connect the starter to a supply voltage of not less than 110 percent of rated voltage at rated frequency. Start and stop a matched motor (hp, voltage, frequency and phases) two times with a 4 minute OFF time between each start. Then all starters shall be energized for not less than 4 consecutive hours. Any malfunction or inability to complete the test shall constitute failure of this test.

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4.5.2.3 Enclosure. Test the enclosure in accordance with NEMA ICS, 1-110, design tests.

4.5.2.4 Fungus. Test the starter in accordance with MIL-STD-810, method 508. Then energize the starter. Evidence of any fungus growth, deterioration, corrosion, or inability to energize at the completion of the fungus test shall constitute failure of this test.

4.5.2.5 Altitude. Expose the starter to an altitude of 6000 feet. Use the starter to start a matched motor (hp, voltage, frequency and phases), with rated voltage at rated frequency. After the motor has accelerated to its rated speed, load the motor to its rated load. The starter shall remain energized and the motor shall run with its rated load for not less than 4 hours. Measure temperature of coils, windings, overload protectors in starter at the time motor is started and at 15 minute intervals until the temperature stabilizes in accordance with NEMA ICS, 1-109.04.8 altitude of not less than 6000 feet. Any malfunction, inability of the starter to energize the motor, overheating above the specified final temperature (ambient temperature plus temperature rise) for the type of insulation used or excessive arcing shall constitute failure of this test. After the 4 hour test period, the starter shall be used to stop the motor running with its rated load. Arcing that does not extinguish shall constitute failure of this test.

4.5.2.6 Dielectric voltage. The dielectric voltage test shall be performed in accordance with NEMA ICS, 1-109-05. Nonconformance to 3.14 shall constitute failure of this test.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Materials, methods, and containers not as specified for level A or B.	Each incorrect material, method or container shall be considered one defect.

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108. Quantities packed together exceed the size or weight limitations of the box for level A or B.
109. Marking missing, illegible, incorrect, or incomplete for level A, B, or C.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each starter shall be preserved in accordance with MIL-P-116, method IC-2. The containers shall be the supplier's standard commercial carton or for each starter not exceeding 10 pounds, the container shall be a close-fitting box conforming to PPP-B-566, variety, style, type, and class optional. For each starter exceeding 10 pounds, the container shall be a close-fitting box conforming to PPP-B-636, type CF, class domestic, variety SW, grade as applicable, style optional. Closure of the boxes shall be in accordance with the appendix to the applicable box specification.

5.1.2 Level C. Each starter shall be preserved and packaged in a manner to assure protection against deterioration and damage from the supplier to the initial destination. The supplier's standard commercial packaging shall be acceptable when it fulfills these requirements.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Starters of like description shall be packed together in close-fitting boxes conforming to PPP-B-601, overseas type, grade B, style optional, or PPP-B-621, class 2, style optional, in quantities not to exceed the weight limitation of the box. Closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B. Starters shall be packed as specified in 5.2.1, except boxes shall be domestic type or class as applicable. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Level C. Starters shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in containers complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

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5.3 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The starters are intended for use in starting 3/4- to 100-hp, alternating-current, induction motors.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, and style required (see 1.2).
(c) Size required (see 3.1).
(d) Time frame required for submission of preproduction model(s) and number of models required (see 3.2).
(e) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct a portion of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2).
(f) Type of enclosure required (see 3.10).
(g) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

6.3 Preproduction model. Any changes or deviations of production starters from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish starters conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and the supplier's maintenance and operation manual to be furnished with each starter.

Custodians:

Army - ME
Navy - YD
Air Force - 11

Review activities:

Army - EL, MI
DSA - GS

User activities:

Army - MU
Navy - MC

Preparing activity:

Army - ME

Project No. 6110-0151

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CENTER
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-S-12514C Starters, Motor; Across-The-Line and Reduced Voltage Types, A-C Induction, 3/4 to 100 HP, Manual and Electric

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426
11 Oct 72 (Rev)

REPLACES EDITION OF 1 OCT 68 WHICH MAY BE USED.

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